

# Corentin Le Guen

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## Senior Backend Developer & Data Engineer

Senior Backend Developer and Technical Lead with a decade of experience in engineering robust, scalable solutions for complex business requirements. Dedicated to leveraging technology to resolve difficult challenges and create tangible value, I am an expert in backend development, real-time data processing, cloud infrastructure, and data analytics. My track record includes modernizing architectures, building mission-critical applications, and deploying state-of-the-art tools for high-frequency trading and financial analysis. Blending technical mastery with strategic vision, I focus on optimizing performance and driving innovation. I thrive in collaborative settings, successfully delivering projects while mentoring teams and fostering talent.

## Technologies

**Back-end:** Python, Rust, Java, Kotlin, Q/KDB+

**Cloud & infrastructure:** Microsoft Azure, AWS, GitHub Actions, Terraform, Ansible

**Front-end:** Angular 20, VueJS, Vite, TypeScript, TailwindCSS, Flowbite, HTML, CSS

**Databases:** Cassandra, SQLite3, MySQL, MSSQL, PostgreSQL, CockroachDB, MongoDB, Redis

**Testing:** Gatling, Mockito, Junit, MockK, Cucumber, Gherkin, Sonar

**Tools & Frameworks:** Tokio, Cargo, Spring Boot, Spring Security, Flask, FastAPI, Prometheus, Grafana, Netlify, Splunk, DataDog, Git, Maven, GitLab / GitHub / BitBucket, RabbitMQ, Postman, PyCharm, IntelliJ, Docker, Jenkins, Flowbite

## Professional Experience

**Alithya**

**since June 2025**

**Senior Backend Developer & Data Engineer**

**Client:** Desjardins.

Technical Advisor for the modernization of a document extraction application.

- Designed and implemented a microservices architecture following current software engineering best practices.
- Developed and modernized Kotlin microservices for file management and data processing.
- Analyzed and decoupled legacy application services to architect a modern microservices-based solution.
- Integrated Figma templates with Angular 20.
- Enhanced application UI/UX to improve user efficiency.
- Automated deployments via GitHub Actions with SonarQube analysis and conducted peer code reviews (Pull Requests).
- Collaborated within a 12-person cross-functional team.

**Stack:** Kotlin | Java | Spring Boot | Spring Security | Hibernates | RabbitMQ | Angular 20 | Redis | IntelliJ | Azure | AKS | Docker | Gradle | JUnit | MockK | Jest | GitHub | GitHub Actions | Sonar | Dynatrace | Splunk

## **Morgan Stanley**

**March 2023 - March 2025**

### **Data Scientist & Strategist**

Served as a technical expert for the trading desk, modernizing legacy systems and streamlining data-driven decision-making processes.

- Architected and maintained advanced data tools and visualization platforms, providing real-time insights and high-availability reporting for trading operations.
- Developed interactive dashboards for trading desks, delivering mission-critical performance metrics and analytics to support high-stakes strategic decisions.
- Collaborated with cross-functional teams to translate business requirements into scalable, data-driven pipelines and architectural solutions.
- Leveraged Q/KDB+ for high-performance, real-time data processing, enabling low-latency high-frequency trading (HFT) strategies and high-throughput data APIs.
- Conducted comprehensive data audits and quality assessments to ensure the integrity, reliability, and regulatory compliance of financial datasets.
- Extracted actionable insights through exploratory data analysis (EDA), providing recommendations to optimize business processes and mitigate operational risks.

**Stack:** Q/KDB+ | Python | Pytest | Docker | BitBucket | TeamCity | Bash | QlikView

## **FX Innovation**

**August 2021 - March 2023**

### **Senior Java Developer & Tech Lead**

**Client:** Desjardins.

Led the technical modernization of mission-critical security and data processing systems, transitioning legacy architectures toward cloud-native, scalable solutions.

- Spearheaded the architectural modernization of Java/Spring APIs dedicated to Oracle security frameworks, ensuring robust authentication and data integrity.
- Engineered and optimized high-throughput microservices for complex file processing, significantly improving system performance and reliability.
- Architected and managed hybrid deployments across on-premises WebLogic servers and Microsoft Azure, bridging the gap during cloud migration.
- Streamlined CI/CD pipelines by migrating workflows from BitBucket/Concourse to GitHub, integrating SonarQube for automated security gates and code quality.
- Championed a culture of quality by implementing Behavior-Driven Development (BDD) using Cucumber and Gherkin, ensuring 100% alignment between technical delivery and business requirements.
- Facilitated cross-team collaboration using Jira and Confluence to document technical roadmaps and mentor junior developers through complex refactoring phases.

**Stack:** Java | Spring Boot | Azure | Docker | Oracle | WebLogic | GitHub Actions | Concourse | Cucumber | SonarQube | Angular | Postman | Jenkins

## **Systematix**

**September 2019 - August 2021**

### **Senior Fullstack Developer**

**Client:** National Bank of Canada.

Led the modernization of financial applications and automated infrastructure management within an OpenShift environment.

- Engineered and maintained scalable Node.js microservices for core banking operations.

- Orchestrated Infrastructure as Code (IaC) using Terraform and Ansible to automate deployments on OpenShift.
- Implemented advanced monitoring and observability solutions using Splunk and DataDog to ensure system uptime.
- Integrated OpenText ExStream for high-volume automated document generation and distribution.
- Optimized CI/CD pipelines using Jenkins and BitBucket, integrating Vault for secure secret management.

**Stack:** NodeJS | FeathersJS | Javascript | Express | Axios | Mocha | OpenShift | Jenkins | BitBucket | Ansible | Nexus | Vault | Jira | Confluence | Bash | Linux | VI | Docker | Dockerfile | Sonar | Splunk | DataDog | PostgreSQL | Postman | Docker | OpenText Exstream | Control-M | MFT

### **Full stack Developer**

**Client:** McGill University.

Developed secure messaging and data validation solutions for academic administration systems.

- Maintained asynchronous messaging systems based on RabbitMQ and Java to ensure reliable data exchange.
- Strengthened API security by implementing authentication and authorization layers within IBM API Connect.
- Developed a custom data-matching algorithm and application to validate complex JSON datasets.

**Stack:** Java | RabbitMQ | Junit | JQuery | API Connect | NodeJS | Javascript | Jira | Confluence | Docker | IBM API Connect | GitLab CI

**Ippon Technologies, France**

**November 2018 - August 2019**

### **Full stack Developer**

**Client:** BNP Paribas.

Contributed to the development of retail banking solutions using Event-Driven Architecture.

- Developed Java/Spring microservices for customer account management within an Agile/Scrum framework.
- Implemented Event-Driven patterns using Apache Kafka (Producers/Consumers) for real-time data streaming.
- Optimized NoSQL data models using Cassandra and Solr to enhance search performance and scalability.
- Executed performance and load testing using Gatling to ensure system resilience under high traffic.
- Mentored junior developers on Java best practices, code quality, and testing methodologies.

**Stack:** Java | Spring | JUnit | Cucumber | Gatling | Kafka | Cassandra | Solr | Git | Jira | Bash | Linux servers

**CGI, France**

**September 2018 - November 2018**

### **Full stack Developer**

**Client:** SFR (Phone operator)

Managed critical legacy system migrations and infrastructure upgrades.

- Spearheaded the migration of ERP systems and Oracle database servers to modernized environments.
- Modernized legacy applications by upgrading core frameworks from Java 6 to Java 8, improving security and performance.
- Maintained professional account management tools, ensuring business continuity during server migrations.

**Stack:** Java 8 | Oracle | Git | Bash

**Altitude Infrastructure, France**

**July 2017 - August 2018**

**Developer**

Designed and maintained business-critical applications for telecommunications management.

- Developed and modernized C#/.NET applications for phone operator management and reporting.
- Built interactive KPI dashboards using Bootstrap and jQuery to provide real-time operational visibility.
- Orchestrated a large-scale ERP migration (from Adonix to Sage X3), including data transcoding and ETL processes.
- Conducted User Acceptance Testing (UAT) to ensure software alignment with business requirements.

**Stack:** C# | .NET | Bootstrap | JQuery | Bash | Git | Redmine | SQL Server | Sage X3 ERP

**Saagie, France**

**October 2016 - June 2017**

**Developer | Intern**

Developed a Big Data analytics solution to track and visualize customer behavior.

- Built a data logging system to track customer journeys using Java and Apache Kafka.
- Developed data pipelines for storage and analysis within Hadoop (Cloudera) and Hive environments.
- Created a custom JavaScript tracking library to capture front-end user interactions seamlessly.

**Stack:** Java | Kafka | Hive | Hadoop | Cloudera | Bash | Javascript | JQuery

## Personal Projects

### Rose

An AWS SDK S3 space usage tracker layer using Rust, CockroachDB, Docker and aws\_sdk\_s3 2025  
[github.com/CorentinLeGuen/rose](https://github.com/CorentinLeGuen/rose)

### Personal Website and portfolio

2025

A personal blog / website using NodeJS, VueJS, TailwindCSS, Flowbite and Netlify  
<https://cleguen.com>

### Project «Life Gamera»

2023

The Conway's Game of Life implementation using Raylib and Python  
<https://github.com/CorentinLeGuen/lifegamera>

### Project «Lotto 6.49 API»

2022

An API containing Lotto draws since 1982-06-12. Python3, Flask, SQLite and Docker  
<https://github.com/CorentinLeGuen/lotto-6-49-api>

### Project «Python on Page»

2022

Running Python on browser using WebAssembly. Python3 and Flask  
<https://github.com/CorentinLeGuen/python-on-page>

### Project «pycasso»

2022

A project to transform a picture into ASCII code using Python and Numpy.  
<https://github.com/CorentinLeGuen/pycasso>

### Renew project «Le Compte est Bon»

2022

Renew the project from 2013 with new technologies to solve the algorithm using Python  
<https://github.com/CorentinLeGuen/le-compte-est-bon>

### Project «array list» in Go

2020

A project made in Go to provide a Java arraylist implementation like  
<https://github.com/CorentinLeGuen/le-compte-est-bon>

### Project «Simple KeyLogger»

2020

A small project to save input keys into a log file. Python3  
<https://github.com/CorentinLeGuen/SimpleKeyLogger>

## Education and Professional Development

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|-----------------------------------------------------------------------------------------------|------|
| <b>Microsoft GitHub Action – certification</b> , Microsoft                                    | 2026 |
| <b>AWS Certified Developer – Associate</b> , Amazon Webservices certificate                   | 2025 |
| <b>Learning AI Through Visualization</b> , Columbia University                                | 2025 |
| <b>Data Science &amp; Machine Learning with Python</b> , Udemy Online course                  | 2022 |
| <b>Certificate of Language Training in English</b> , ACA Online course at lesateliers.ca      | 2021 |
| <b>Specialization: Programming with Google Go</b> , Coursera Online course                    | 2019 |
| <b>Intermediate R Course</b> , DataCamp Online course                                         | 2019 |
| <b>Deep Learning</b> , National Conservatory of «Arts et Métiers», France Online course       | 2018 |
| <b>Master degree of Computer Software Engineering</b> , University of Rouen Normandie, France | 2018 |
| <b>Licence Diploma of Development</b> , University of Rouen Normandie, France                 | 2016 |
| <b>Introduction to C++ Development</b> , Polytechnic School of Lausanne, Swiss                | 2013 |
| <b>Introduction to Java Development</b> , Polytechnic School of Lausanne, Swiss               | 2013 |

### Languages

French - Native  
English – Fluent (C1 level)  
Dutch - Elementary  
Russian - Elementary